Fusing Current Checker (FCC) per wire

Est. CCC: Original FCC vs Improved FCC (Copy Version)

Item	Original Version	Improved Version	Why Improvement Needed
CCC Method	Fixed ΔT = 390°C for all cases	Multiple ΔT models based on condition	More realistic representation of wire behavior
Temperature Input	Not considered	User can enter Operating Temperature	Different products have different operating temperatures
Product Specificity	Same assumption for all products	Product-dependent calculation	Better reflects actual application
Design Evaluation	RT CCC only	Design CCC	Standard benchmark for comparison
Real Product Evaluation	Not available	Operating CCC	Main value used for wire-size selection
Thermal Stress Evaluation	HT CCC only	Extreme Stress CCC	Better understanding of wire robustness
Operating Temperature Guidance	None	PRS / CRS / SSLT reference	Helps user select appropriate temperature
Wire Length Sensitivity	Single value	CCC Range (Min~Max)	Accounts for wire-length variation
User Understanding	RT / HT terminology	Design / Operating / Extreme	Easier for non-experts to understand
Design Decision	Difficult to determine which value to use	Operating CCC clearly identified	Reduces misuse of results

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Est. CCC: What Do The Three CCC Values Mean?

CCC Type	Purpose	Temperature Model	When To Use
Design CCC	Standard benchmark comparison	25°C → 150°C	Compare wire materials and diameters
Operating CCC	Real product condition	Operating Temp → Wire Tmax	Final wire-size selection ✅
Extreme Stress CCC	Thermal robustness comparison	50% MP → 80% MP	Stress/reference study only

Simple Non-Engineering Explanation

Think of a bond wire like a road.

Condition	Simple Explanation
Design CCC	How many cars can the road carry under a standard test?
Operating CCC	How many cars can the road carry during everyday traffic?
Extreme Stress CCC	How many cars can the road carry during an emergency situation?

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Est. CCC: Why Operating CCC Can Be Lower Than Extreme Stress CCC

- Operating CCC is calculated using the actual product operating temperature, while Extreme Stress CCC is calculated using a much larger allowable temperature-rise range.
- Because Extreme Stress CCC allows a greater temperature increase before reaching its defined limit, the calculated current capacity may sometimes be higher than Operating CCC.
- This is expected behavior and does not mean the wire should be operated at the Extreme Stress CCC current.

Which Value Should Be Used?

Purpose	Value to Use
Wire-Size Selection	✅ Operating CCC
Continuous Operation	✅ Operating CCC
Design Comparison	✅ Design CCC
Thermal Robustness Study	✅ Extreme Stress CCC
Failure Reference	✅ Fusing Current